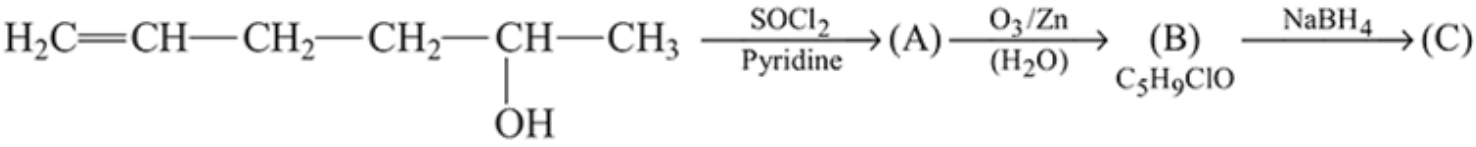
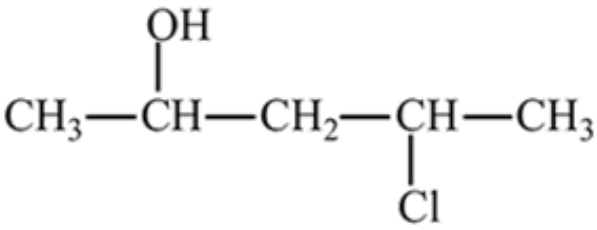
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1



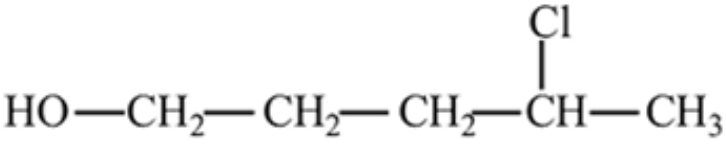
Compound (C) is

(A)

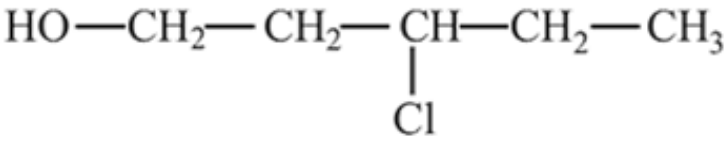


(B) $HOCH_2 - CH_2 - CH_2 - CH_2 - CH_2 - CH_2 - Cl$

(C)

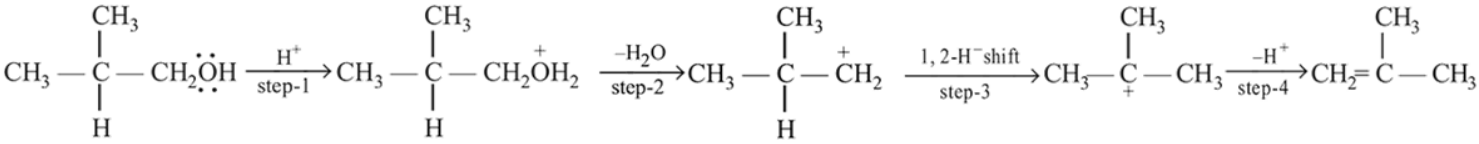


(D)



Q.2

Dehydration of alcohols by conc. H_2SO_4 takes place according to following steps:



The slowest and fastest steps in the above reaction are

(A) step 1 is slowest, while 3 is fastest.

(B) step 2 is slowest, while 3 is fastest.

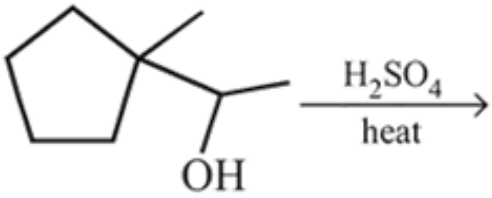
(C) step 2 is slowest, while 4 is fastest.

(D) all steps proceed at equal rate.

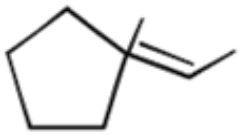
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Q.3

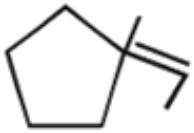
The most probable product in the reacion given below is



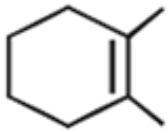
(A)



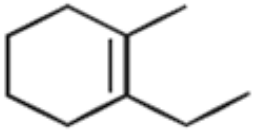
(B)



(C)



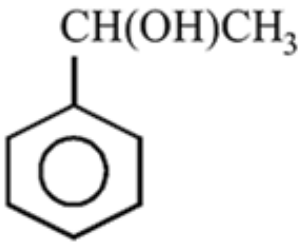
(D)



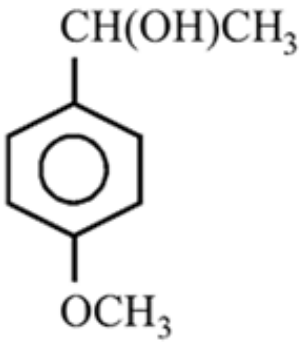
Q.4

Which of the following alcohols is dehydrated most readily with conc. H_2SO_4?

(A)

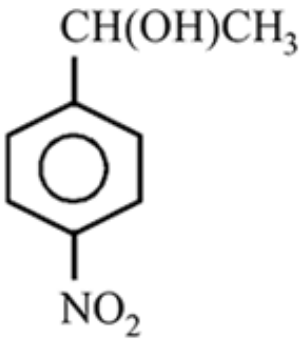


(B)

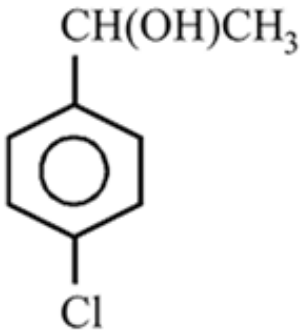


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(C)



(D)



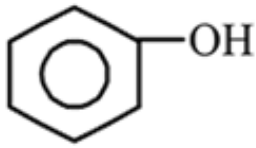
Q.5

Which one of the following compounds has the most acidic nature?

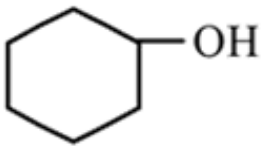
(A)



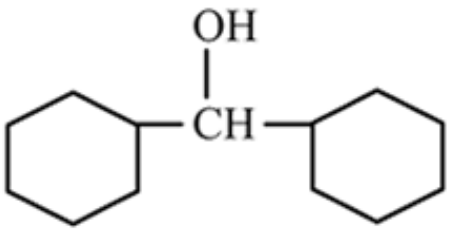
(B)



(C)



(D)



Q.6

The major product P in the following reaction is

$$(\mathrm{CH_3})_3\,\mathrm{COH} + \mathrm{C_2H_5OH} \xrightarrow{\mathrm{H^+}} \mathrm{P}$$

(A) $(\mathrm{CH_3})_3\,\mathrm{COC}\,(\mathrm{CH_3})_3$

(B) $(\mathrm{CH_3})_3\,\mathrm{COC_2H_5}$

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(C) $C_2H_5OC_2H_5$

(D) $(CH_3)_2C=CH_2$

Q.7

Correct increasing order of acidity is as follows

(A) H_2O, C_2H_2, H_2CO_3, phenol

(B) C_2H_2, H_2O, H_2CO_3, phenol

(C) phenol, C_2H_2, H_2CO_3, H_2O

(D) C_2H_2, H_2O, phenol, H_2CO_3

Q.8

What is Z in the following sequence of reactions?

$$Z \xrightarrow{PCl_5} X \xrightarrow{\text{Alc. KOH}} Y \xrightarrow[(ii)\,H_2O;\ \text{boil}]{(i)\,\text{Conc. } H_2SO_4} Z$$

(A) $CH_3CH_2CH_2OH$

(B) $CH_3CH(OH)CH_3$

(C) $(CH_3CH_2)_2CHOH$

(D) $CH_3CH=CH_2$

Q.9

Arrange the following in increasing order of their acidity?
(a) o -cresol, (b) salicylic acid, (c) phenol

(A) $c < a < b$

(B) $b < c < a$

(C) $a < b < a$

(D) $a < c < b$

Q.10

An organic compound $\mathbf{A}$ reacts with methyl magnesium iodide to form an addition product which on hydrolysis forms the compound $\mathbf{B}$. Compound $\mathbf{B}$ gives blue colour salt in Victor Meyer's test. The compounds $\mathbf{A}$ and $\mathbf{B}$ are respectively

(A) acetaldehyde, tertiary butyl alcohol

(B) acetaldehyde, ethyl alcohol

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(C) acetaldehyde, isopropyl alcohol

(D) acetone, isopropyl alcohol

Q.11

The strongest acid among the following aromatic compounds is

(A) p -chlorophenol

(B) p -nitrophenol

(C) m-nitrophenol

(D) O-nitrophenol

Q.12

Which compound is formed when CH_3OH reacts with $CH_3 - Mg - X$

(A) Acetone

(B) Alcohol

(C) Methane

(D) Ethane

Q.13

Methyl alcohol can be distinguished from ethyl alcohol using

(A) Fehling solution

(B) Schiff's reagent

(C) Sodium hydroxide and iodine

(D) Phthalein fusion test

Q.14

The reagent which easily reacts with ethanol and propanol is

(A) Fehling solution

(B) Grignard reagent

(C) Schiff's reagent

(D) Tollen's reagent

Q.15

$CH_3CH_2OH \rightarrow CH_3CH_2Cl$

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Find out number of reagents that can be used for above transformation from following
$HCl/ZnCl_2, P + Cl_2/\Delta, SOCl_2, TsCl, NaCl, PCl_3, PCl_5, AgCl$

(A) 4

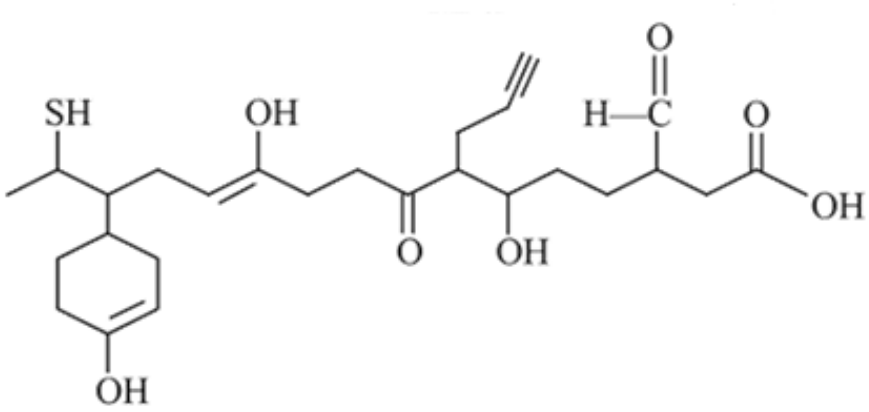
(B) 5

(C) 7

(D) 9

Q.16

Find number of moles of CH_4 obtained when 1 mole of the given compound reacts with excess of CH_3MgBr



(A) 7

(B) 9

(C) 6

(D) 3

Q.17

An alcohol on oxidation is found to give CH_3COOH and CH_3CH_2COOH. The structure of the alcohol is

(A) $CH_3CH_2CH_2OH$

(B) $(CH_3)_2 C(OH)CH_2CH_3$

(C) $CH_3CH_2CHOHCH_3$

(D) $CH_3CH(OH)CH_2CH_2CH_3$

Q.18

Dehydration of cyclopentyl carbocation with conc. H_2SO_4 forms

(A) Cyclopentene

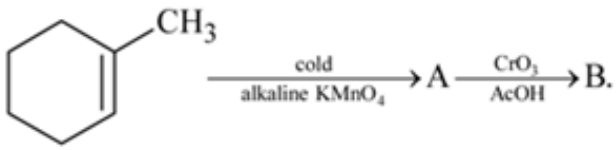
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(B) Cyclohexene

(C) Cyclohexane

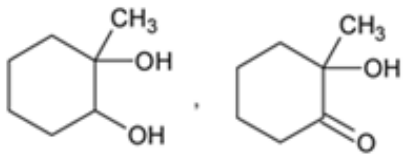
(D) None of these

Q.19

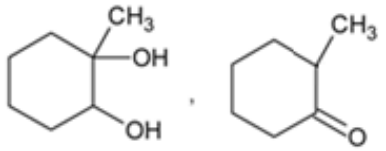


A and B are

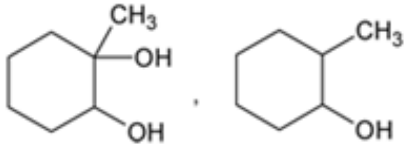
(A)



(B)



(C)

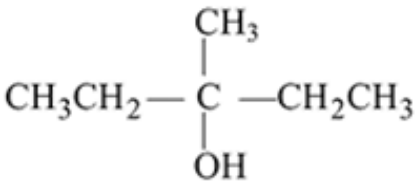


(D) No formation of A and B

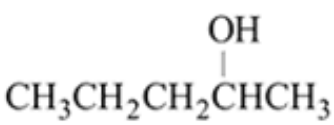
Q.20

Among the following compounds which can be dehydrated very easily is

(A)

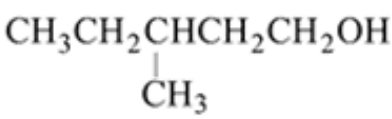


(B)



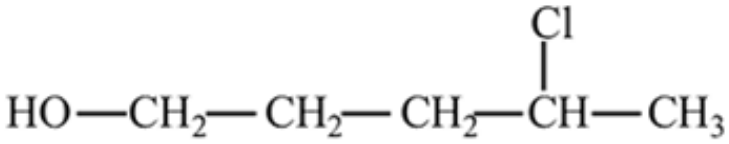
(C) $CH_3CH_2CH_2CH_2CH_2OH$

(D)

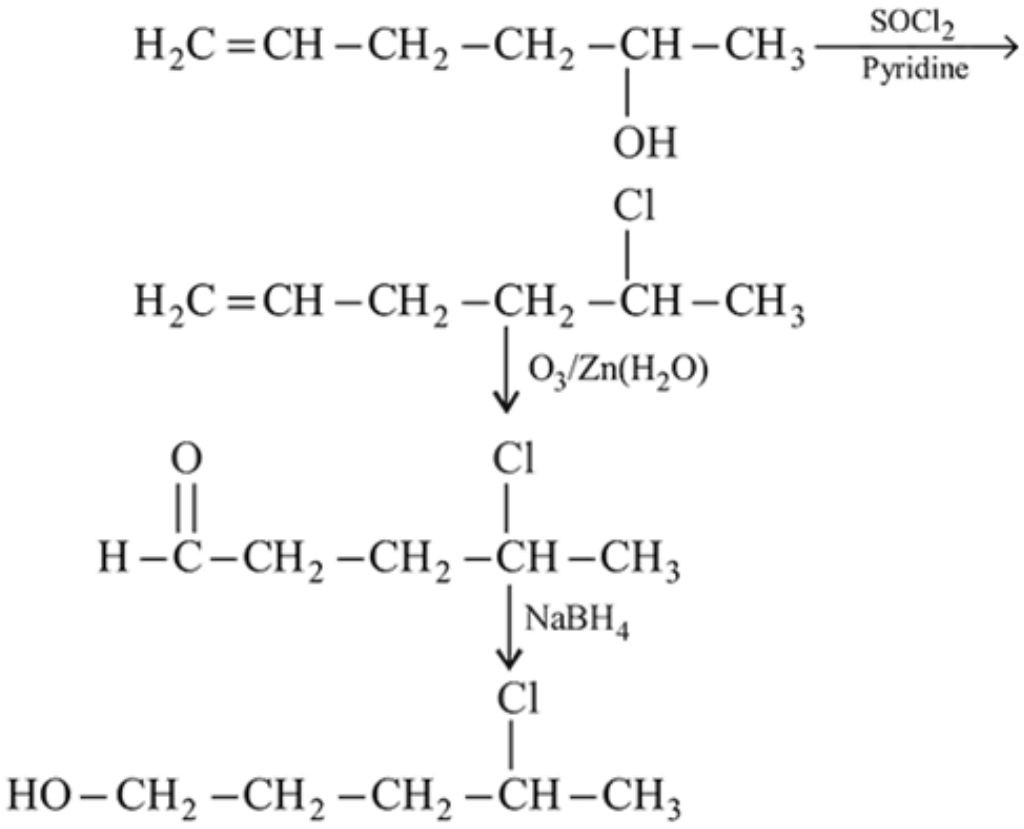


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Q.1 Answer:



Solution:



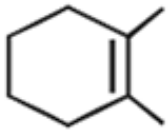
Q.2 Answer:

step 2 is slowest, while 4 is fastest.

Solution:

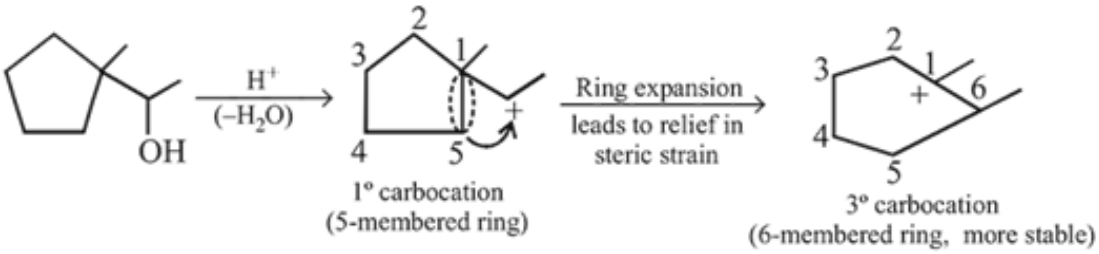
Step 2 involves the formation of carbonium ion by the loss of weakly basic H_2O molecule. It is slowest step. Step 4 involves the conversion of an unstable (or intermediate) into a quite stable product, hence it is fastest step.

Q.3 Answer:

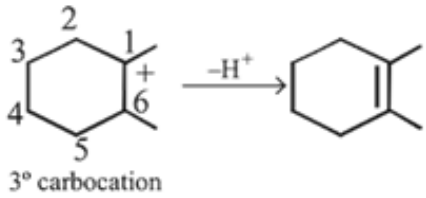


Solution:

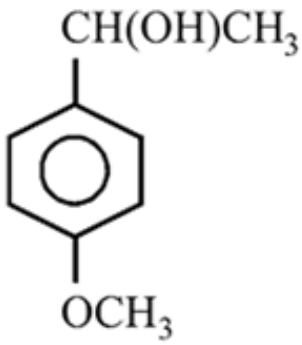
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or

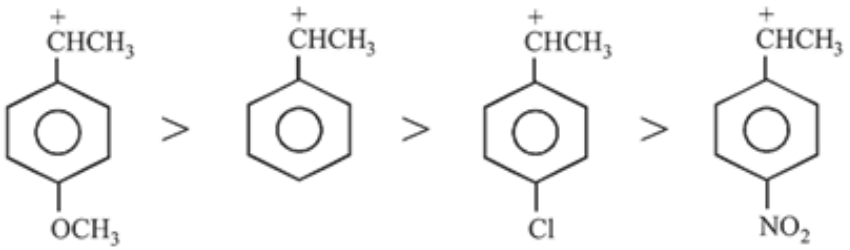


Q.4 Answer:

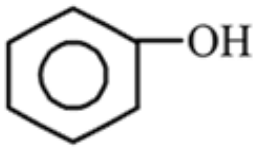


Solution:

The reaction involves the formation of carbocation as intermediate. Hence more the stability of the carbocation, more will be the rate of reaction. Let us draw the structure of the corresponding carbocation and observe the relative stability of the four benzyl carbocations. The relative stability of the four carbocations is



Q.5 Answer:



Solution:

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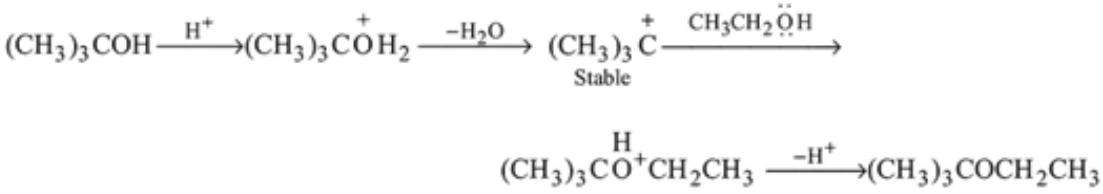
Phenol is most acidic because its conjugate base is stabilised due to resonance, while the rest three compounds are alcohols, hence, their corrosponding conjugate bases do not exhibit resonance.

Q.6 Answer:

$(CH_3)_3 COC_2H_5$

Solution:

tert - Butyl alcohol is preferentially protonated because its corresponding carbocation is highly stable than that of C_2H_5OH



The tert-butyl cation is preferentially attacked by CH_3CH_2OH rather than bulky $(CH_3)_3 COH$

Q.7 Answer:

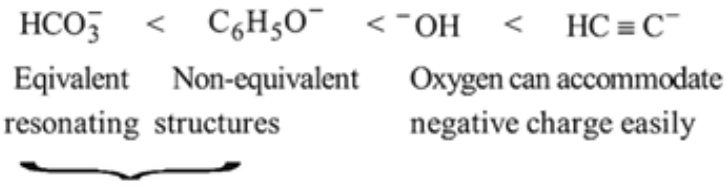
C_2H_2, H_2O, phenol, H_2CO_3

Solution:

Such questions can be solved by considering the relative basic character of their conjugated bases which for H_2O, C_2H_2, H_2CO_3 and C_6H_5OH are

$^-OH, HC \equiv C^-, HCO_3^-, C_6H_5O^-$

More the possibility for the dispersal of the negative charge, weaker will be the base. Thus the relative basic character of the four bases is



Due to resonance

Thus the acidic character of the four corresponding acids will be $H_2CO_3 > C_6H_5OH > H_2O > HC \equiv CH$

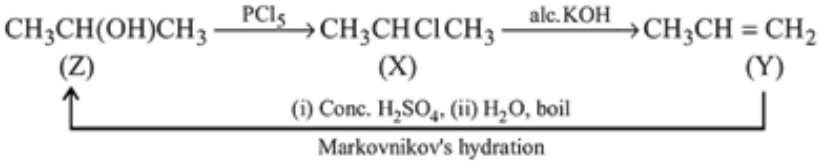
Q.8 Answer:

$CH_3CH(OH)CH_3$

Solution:

Reagents used in the various steps indicate that the compound Z has an alcoholic group. This set of reactions is possible only when Z is $CH_3CHOHCH_3$.

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In options a and c, Y cannot be converted back into d by the given series of reactions.

Q.9 Answer:

$a < c < b$

Solution:

Electron releasing groups etc) intensify the negative charge of phenoxide ion, i.e., destabilizes it hence decreases ionization of parent phenol, therefore decreases acidity. While electron withdrawing groups etc. increases acidity. $\mathrm{CH_3}$ is an electron-donating group so its acidity is the least. Salicylic acid has two electron-donating groups so its acidity is the highest.

Therefore, the correct order of increasing order of acidity is:

$a < c < b$ |

Q.10 Answer:

acetaldehyde, isopropyl alcohol

Solution:

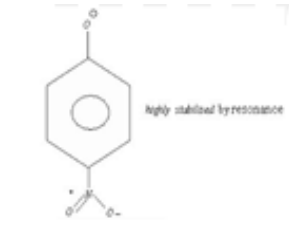
Compound B is a Sec. alcohol since it gives blue colour in Victor-Meyer test and Sec alcohol are obtained by the action of $\mathrm{CH_3MgI}$ on an aldehyde other than formaldehyde.

So A is Acetaldehyde and B is Isopropyl Alcohol

Q.11 Answer:

p -nitrophenol

Solution:



Nitro group is strongly electron withdrawing group. At ortho position intra-molecular hydrogen bond reduces the acidic strength.

Q.12 Answer:

Methane

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Solution:

$ROH + R^1mg \rightarrow R^14 + ROMg$

$CH_3 - OH + CH_3MgX \text{ ®} CH_4 + CH_3O - MgX$

Q.13 Answer:

Sodium hydroxide and iodine

Solution:

Ethyl Alcohol gives iodoform test

$C_2H_5OH \xrightarrow[\text{Iodoform Teste}]{NaOH/1_2} CHI_3$ **(yellow ppt)**

$CH_3OH \xrightarrow{NaOH/L_2}$ **No ppt.**

Q.14 Answer:

Grignard reagent

Solution:

$C_2H_5OH + R - Mg - X \text{ ®} RH + C_2H_2OMgX$

$C_3H_7OH - R - Mg - X \text{ ®} RH + C_3H_7OMgX$

Q.15 Answer:

4

Solution:

$P/Cl_2, SOCl_2, PCl_3$ **and** PCl_5.

Q.16 Answer:

6

Solution:

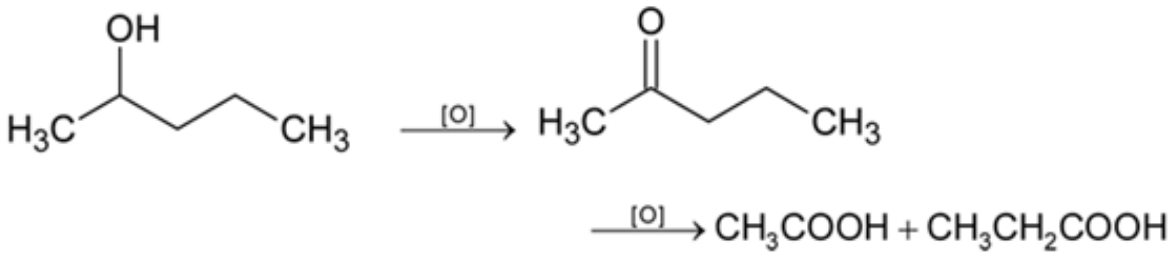
All OH, SH and terminal alkyne will react with Grignard reagent to produce CH_4.

Q.17 Answer:

$CH_3CH(OH)CH_2CH_2CH_3$

Solution:

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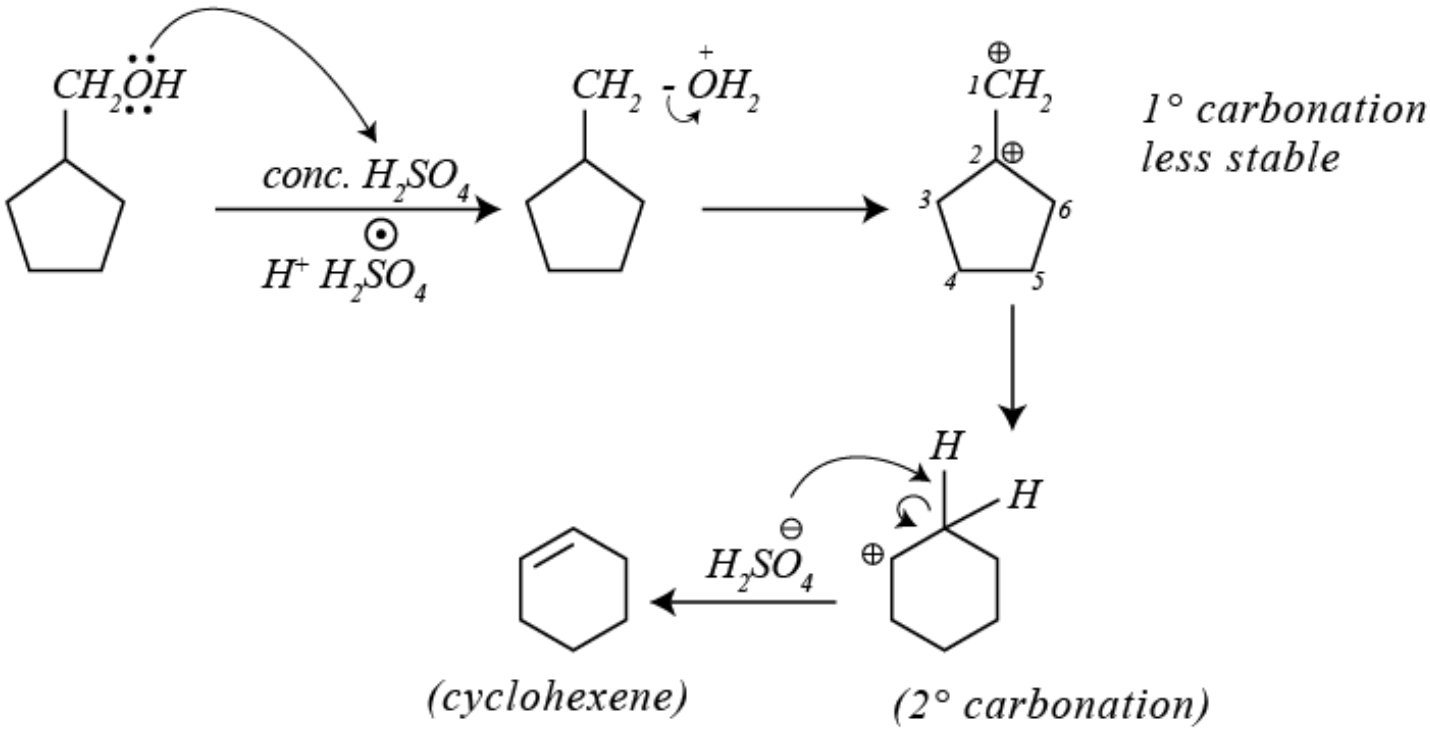


Q.18 Answer:

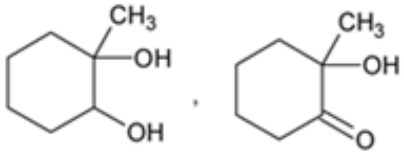
Cyclohexene

Solution:

Dehydration of cyclopentyl carbinol with conc. H_2SO_4 forms a cyclic compound. Write the structure of the product.



Q.19 Answer:

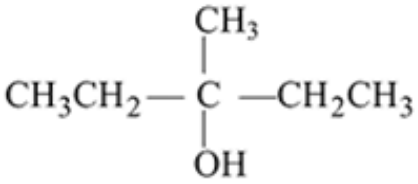


Solution:

2° alcohol oxidizes but 3° does not.

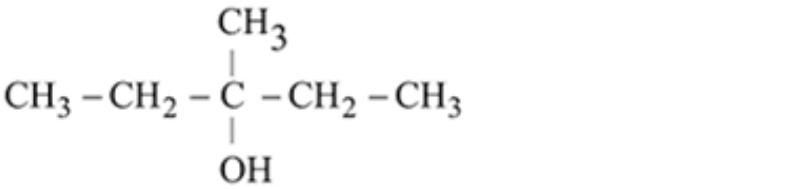
Q.20 Answer:

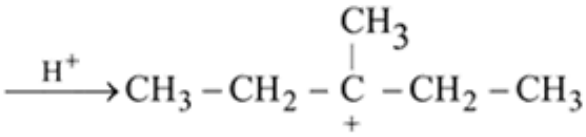
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Solution:

3-Methyl pentanol-3 will be dehydrated most readily since it produces tertiary carbonium ion as intermediate.





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